Calcurion Physics Worksheets

A-Level Physics

Topic: Mechanics & Materials

MARKSCHEME

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1. $F = \Delta p/\Delta t$ (rate of change of momentum) **A2** [2]
2. $\Delta p = 6$; $v = 2$ m/s **M1A1** [3]
3. Turning effect of a force about a point **A1** [1]
4. 10 Nm **M1A1** [2]
5. Sum of clockwise moments = sum of anticlockwise moments **A2** [2]
6. $\rho = m/V$; kg m^{-3} **A2** [2]
7. 4000 kg m^{-3} **M1A1** [2]
8. $F = k\Delta L$ (within limit of proportionality) **A2** [2]
9. $F = 6$ N; $E = 0.12$ J **M1A1** [4]
10. Stress $= F/A$; strain $= \Delta L/L$ **A2** [2]
11. 2.0×10^7 Pa **M1A1** [2]
12. Stress / strain (in elastic region) **A1** [1]
13. $E = 1.6 \times 10^{11}$ Pa **M1A1** [3]
14. Maximum stress for which material returns to original length **A1** [1]
15. Linear region then curve; UTS at maximum stress **A3** [3]
16. $v = 1.6$ m/s **M1A1** [3]
17. $T = 60$ N **M1A1** [3]
18. Area under F–ΔL graph $= \frac{1}{2}F\Delta L$ **A3** [3]
19. $A = 5.03 \times 10^{-7}$ m^2; $\Delta L = 0.75$ mm **M1A1** [5]
20. Elastic: returns to original shape; plastic: permanent deformation **A3** [3]
21. $A = 7.85 \times 10^{-7}$ m^2; $F_{\max} = 314$ N **M1A1** [3]
22. $a = -6.25$ m/s^2; $F = 7500$ N **M1A1** [3]
23. Resultant force zero; resultant moment zero **A2** [2]
24. $k = 392$ N/m; $E = 0.49$ J **M1A1** [4]
25. Measure L, d, F, ΔL; plot stress–strain; gradient $= E$ **A5** [5]